



Unit-5 - Hydrogen and Fuel Cell Technology

HYDROGEN - PROPERTIES - PRODUCTION TECHNIQUES - STORAGE - TRANSPORTATION. FUEL CELL PRINCIPLES AND ITS CLASSIFICATION, TYPES - PHOSPHORIC ACID, POLYMER ELECTROLYTE MEMBRANE FUEL CELL, MOLTEN CARBONATE FUEL CELL AND SOLID OXIDE FUEL CELL, FUEL CELL CONVERSION EFFICIENCY AND APPLICATIONS.

Hydrogen

- Hydrogen is a colorless, odorless, highly flammable gas with low density, produced via electrolysis, steam reforming, and other methods.
- It is stored as high-pressure gas, cryogenic liquid, or chemically in materials like metal hydrides.
- Transportation occurs via pipelines for large volumes, high-pressure or cryogenic trucks, ships, or trains, with the method chosen based on scale and economics.

Properties

(a) Physical properties:

Appearance: Colorless, odorless, and tasteless gas.

Density: Lighter than air; the lowest density of any gas.

Boiling/Freezing Point: Extremely low, boiling at -253°C and freezing at 259°C .

Flammability: Highly flammable, with a wide flammability range (4–75% in air).

Flame characteristics: The flame is pale blue and almost invisible in daylight, making leak detection difficult without sensors.

Diffusivity: As a very small molecule, hydrogen has a high diffusion rate, making it prone to leakage.

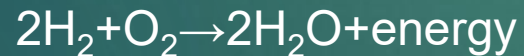
Properties

(b) Chemical properties:

High energy content: Contains the highest energy per unit of mass among all fuels (120 MJ/kg).

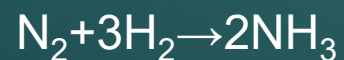
Reactivity: Relatively stable at room temperature but highly reactive at elevated temperatures or when ignited.

Combustion: Burns in air or oxygen to form water, releasing a large amount of energy.



Reducing agent: Can reduce metal oxides to their pure metal form.

Reaction with nitrogen: Reacts with nitrogen in the Haber process to form ammonia



Comparison of various properties of hydrogen with other fuels

Property	Gasoline	Natural Gas	Hydrogen
Density (kg/m ³)	730	0.78	0.0837
Boiling Point (°C)	38 – 204	–156	–253 (20.3 K)
Lower Heating Value (MJ/kg)	44.5	48	125
LHV (MJ/m ³)	32	37.3	10.4 (gas), 8520 (liq.)
Higher Heating Value (MJ/kg)	50.8	55	141.9
HHV (MJ/m ³)	36.6	42.6	11.89 (gas), 10046 (liq.)
Flammability Range (% in air)	1.4 – 7.6	5 – 16	4 – 75
Flame Speed (m/s)	0.4	0.41	3.45
Flame Temperature (°C)	2197	1875	2045
Flame Luminosity	High	Medium	Low

Production techniques

(a) Electrolytic processes :

These use electricity to split water into hydrogen and oxygen



❑ Polymer Electrolyte Membrane (PEM) Electrolysis:

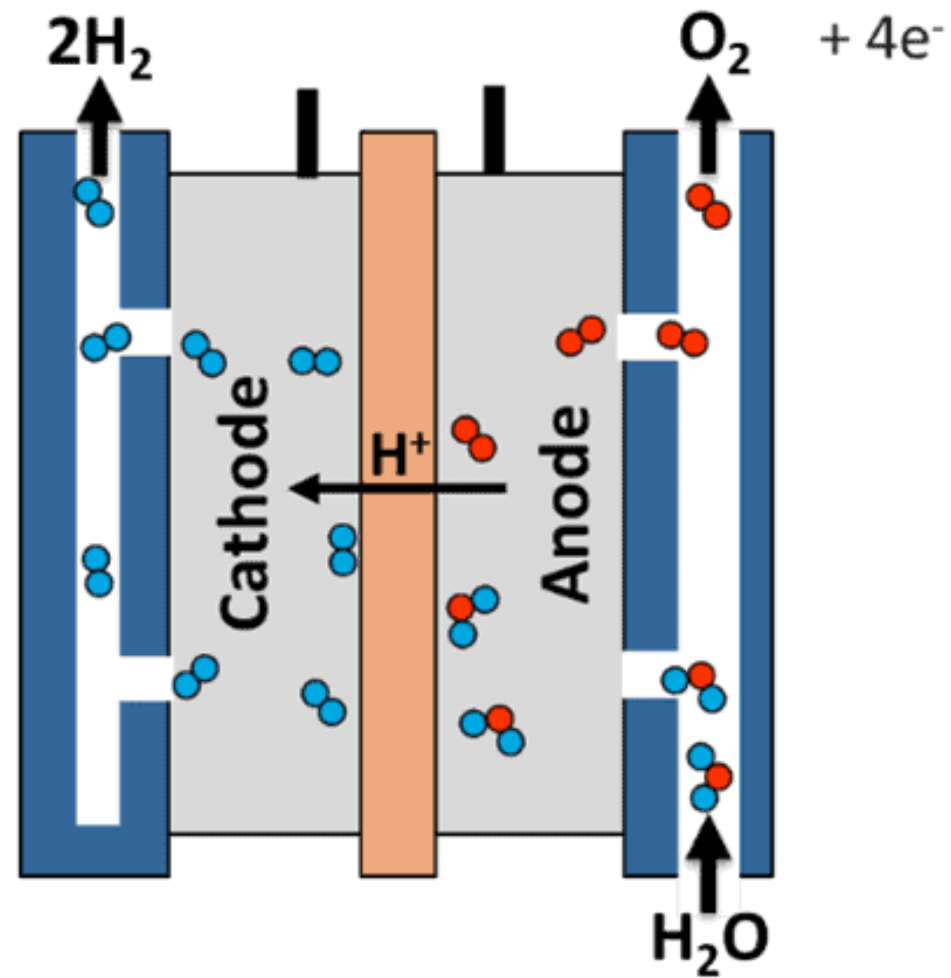
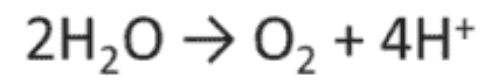
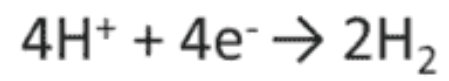
- Uses a **solid polymer membrane** to separate the hydrogen and oxygen.
- It is **highly efficient**, with a fast response time, making it ideal for pairing with variable renewable energy sources.
- **most widely used.**

❑ Alkaline Electrolysis (AEC):

- A mature and reliable technology that uses a **liquid alkaline electrolyte**, such as potassium hydroxide.
- It has a **longer lifespan and lower capital costs** than PEM technology.

❑ Solid Oxide Electrolysis (SOEC):

- **Operates at very high temperatures and efficiencies, producing hydrogen from steam.**

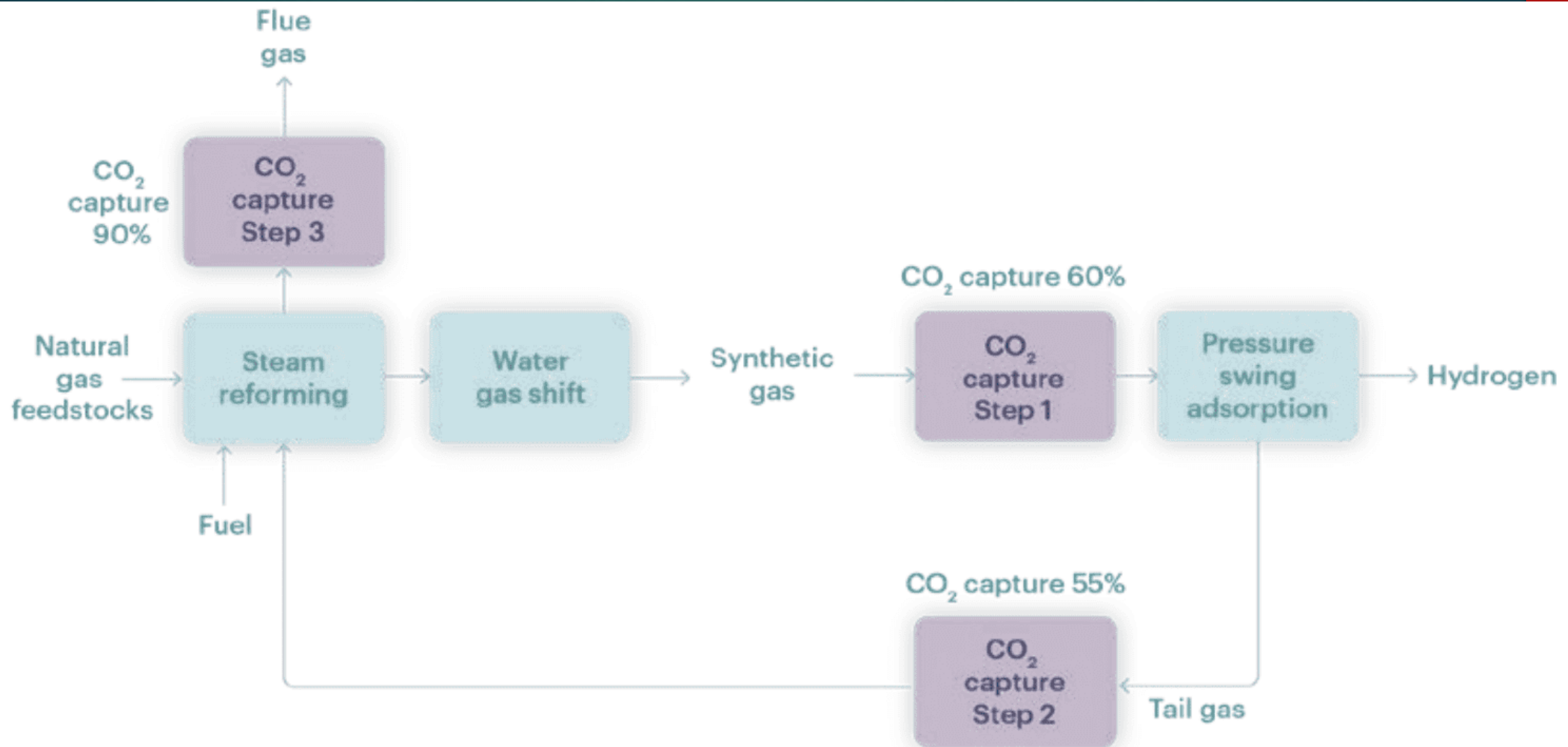


Production techniques

(b) Thermochemical processes

These methods use heat and chemical reactions to extract hydrogen from carbon-based or water-based sources.

- **Steam Methane Reforming (SMR):** The most common and established method, involving high-temperature steam reacting with methane (natural gas) to produce syngas (H_2 and CO). Carbon dioxide is a byproduct.
- **Process Steps:**
 - Reforming Reaction:**
 - Methane (CH_4) + Steam (H_2O) $\rightarrow$ Hydrogen (H_2) + Carbon Monoxide (CO) + Carbon Dioxide (CO_2).
 - Water-Gas Shift Reaction: $CO + H_2O \rightarrow CO_2 + H_2$ (catalyst-assisted, increases H_2 yield).
 - Purification (PSA – Pressure Swing Absorption): Removes CO_2 and impurities $\rightarrow$ pure hydrogen.



Source: IEAGHG (2017a), "Reference data and supporting literature reviews for SMR based hydrogen production with CCS".

(C) Biomass Gasification:

Converts **biomass**, such as agricultural waste or wood, **into syngas** through high-temperature reactions with a gasifying agent like steam, oxygen, or air.

Process Steps:

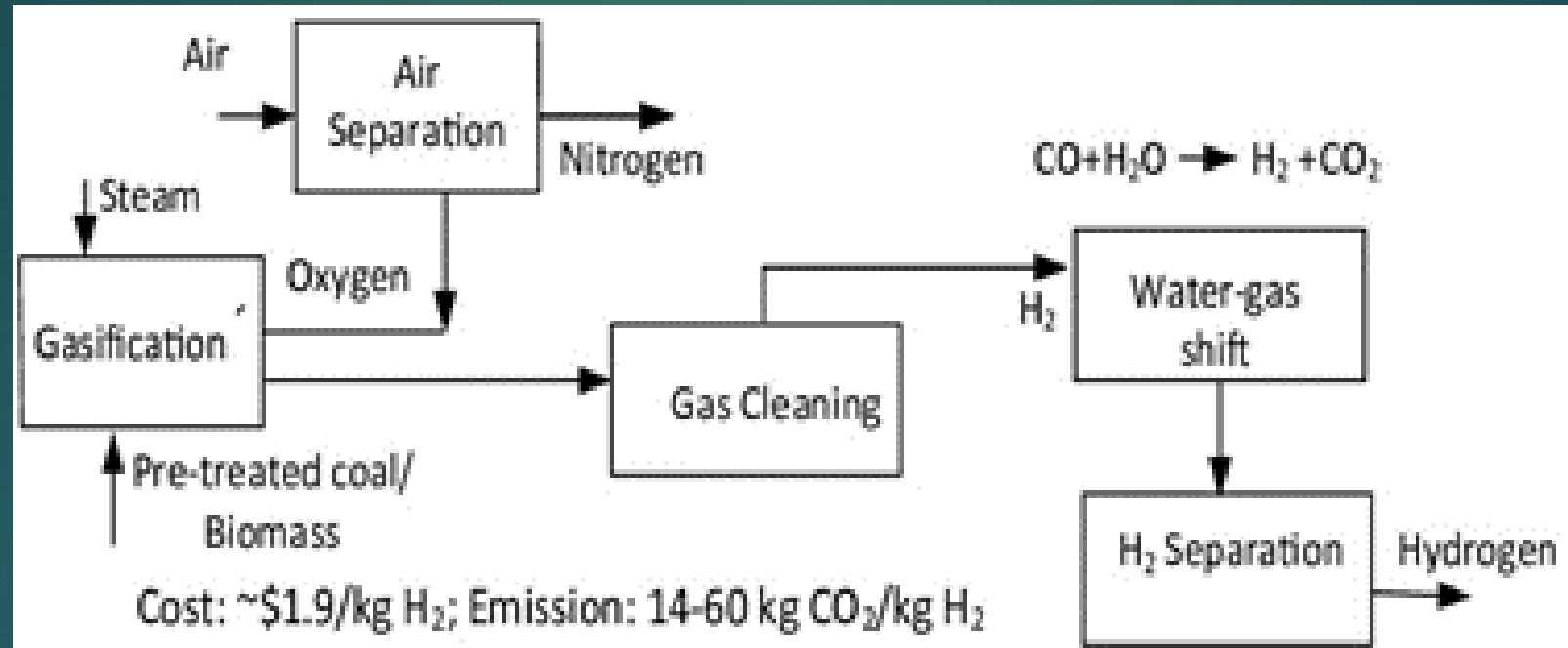
Gasification: $\text{Coal} + \text{Steam} \rightarrow \text{Synthesis Gas (CO, CO}_2, \text{H}_2)$.

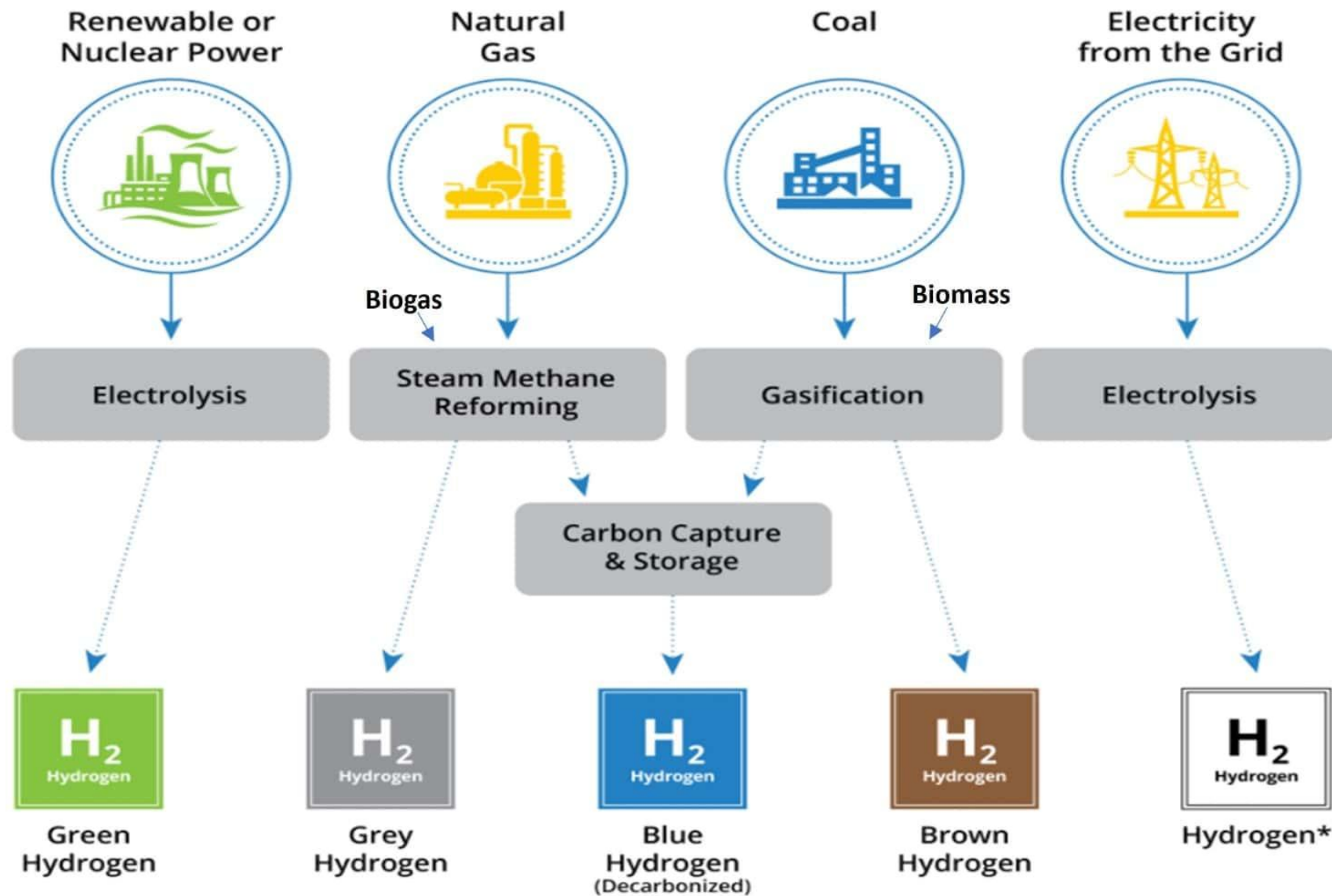
Water-Gas Shift Reaction: $\text{CO} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + \text{H}_2$ (increases hydrogen yield).

Purification: Separation of H_2 from CO_2 and other gases.

Carbon Emissions: Gasification emits **2–3× more CO_2** compared to steam methane reforming (SMR).

Gasification





**Emissions depend on the mix of electricity sources on the grid*

Green Hydrogen: Produced when electricity is from renewables (solar, wind) or nuclear (zero-carbon).

Non-Green Hydrogen: Produced when electricity is from fossil fuels (coal, oil, gas), resulting in high CO₂ emissions.

Hydrogen Storage Methods

1. Physical Storage:

- **Compressed Gas:** High-pressure tanks (350–700 bar); mature tech for fuel cells and industry.
- **Liquid Hydrogen:** Cooled to $-253\text{ }^{\circ}\text{C}$ in insulated tanks; high volumetric energy density but energy-intensive and prone to boil-off.
- **Cryo-compressed Hydrogen:** Combines low temperature and high pressure; higher energy density and reduced boil-off.

2. Material-Based Storage:

- **Metal Hydrides:** Hydrogen stored chemically in metals; safer but heavy; requires heat for release.
- **Liquid Organic Hydrogen Carriers (LOHCs):** Hydrogen bonded to organic liquids; released via catalytic dehydrogenation.
- **Ammonia (NH_3):** Liquid at moderate temperatures; hydrogen released via energy-intensive cracking.
- **Underground Storage:** Large-scale storage in salt caverns or depleted fields.

Transportation

Hydrogen can be transported from production sites to end-users via several methods, each with different cost, safety, and efficiency profiles.

Pipeline: For large volumes over long distances, pipelines are a cost-effective option. Existing natural gas pipelines can be repurposed for hydrogen, though this may require modifications to prevent hydrogen embrittlement and leakage.

Truck: Hydrogen can be transported via road in either gaseous or liquid form.

Tube Trailers: Carry compressed gaseous hydrogen in high-pressure cylinders. This is best suited for shorter distances due to limited capacity per truck.

Cryogenic Tanker Trucks: Transport liquid hydrogen. Though more efficient for larger volumes, it requires special insulation and faces boil-off losses.

Ship: For international or long-distance transport, hydrogen can be shipped as a liquid or bound to a carrier, such as ammonia. Shipping as ammonia is more efficient, but requires cracking technology at the destination.

Hydrogen Carrier Materials: Transporting hydrogen in chemical carriers like LOHCs or ammonia can simplify logistics by allowing use of existing infrastructure, but require energy for release.

FUEL CELL

Definition: A fuel cell is an electrochemical device that continuously converts the chemical energy of a fuel directly into electrical energy.

Operation:

- Continuous supply of **fuel** and **oxidant** required.
- Removal of **water vapour, heat, and spent gases** needed for operation.
- Contains **two electrodes** (anode and cathode) with an electrolyte in between.
- **Anode (fuel electrode):** Fuel (e.g., H_2) is supplied.
- **Cathode (oxidant electrode):** Oxidant (e.g., O_2 or air) is supplied.
- Produces **DC power** directly (static conversion).

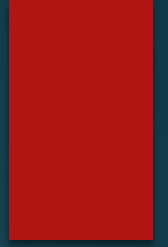
By-products:

- If pure H_2 and O_2 are used → only **water vapour**.
- If hydrocarbon fuel → **CO_2** and **water vapour**.
- Using air as oxidant → **nitrogen** (spent air) also in exhaust.

Efficiency:

- Around 50% (higher than thermal systems).
- Not limited by Carnot efficiency since no intermediate thermal stage.
- Voltage: ~ 0.7 V per cell; cells are connected in series/parallel for desired output.

Advantages of Fuel Cells

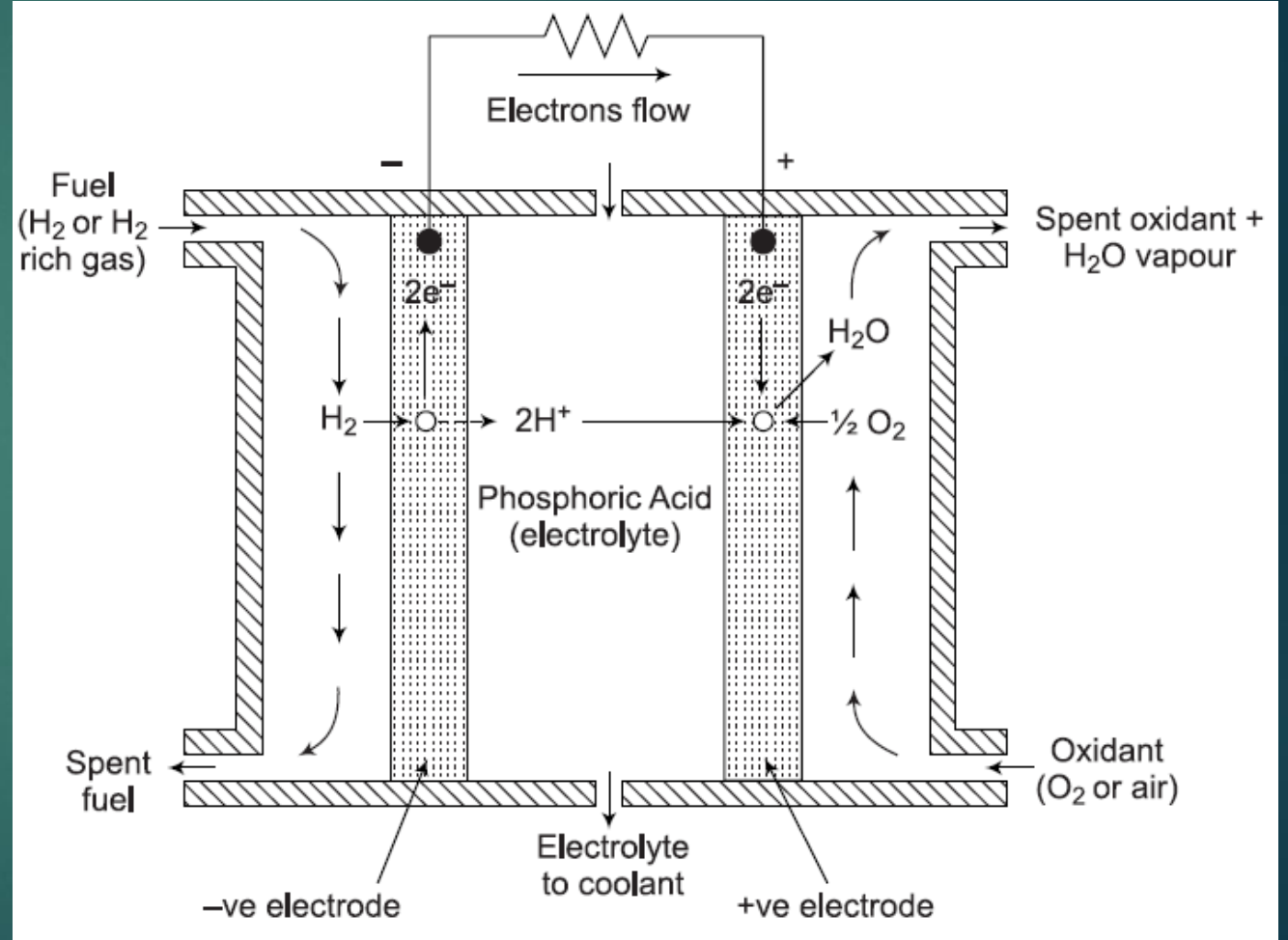


- **Quiet Operation:** Static device with no moving parts.
- **Eco-friendly:** Minimal pollutants—mainly water and CO₂.
- **High Efficiency:** Direct chemical-to-electrical energy conversion.
- **Decentralized Installation:** Can be located near point of use; avoids transmission losses.
- **No Cooling Water Needed:** Waste heat can be used for heating.
- **Modular Design:** Easy to scale voltage/current as required.
- **Compact & Space-efficient:** Ideal for both stationary and mobile use.
- **Fuel Flexibility:** Can use hydrogen, hydrocarbons, or other fuels.
- **Good Part-load Efficiency:** Operates efficiently from 50–100% load.
- **No Charging Required:** Continuous operation with steady fuel supply.

Classification of Fuel Cells

1. Phosphoric Acid Fuel Cell (PAFC)

Uses phosphoric acid as the electrolyte. These cells operate at higher temperatures than AFCs and can use a wider variety of fuels, though they are still sensitive to impurities.



1. Phosphoric Acid Fuel Cell (PAFC)

Developed in **1980s**; uses **phosphoric acid** as electrolyte.

Electrodes: Porous, conductive material (typically **nickel**) with **catalyst** coating.

Fuel: Pure hydrogen or H₂-rich gas.

Oxidant: Oxygen or air.

Catalyst: **Platinum (Pt)** — best but costly; **Ni** (anode) and **Ag** (cathode) also used.

Catalyst Poisoning: Caused by **impurities (e.g., sulfur compounds)** reducing activity.

Electrochemical Reactions:

Anode: $\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$

Cathode: $\frac{1}{2}\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2\text{O}$

Overall: $\text{H}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{H}_2\text{O} + \text{Electrical Energy}$

Operating Conditions:

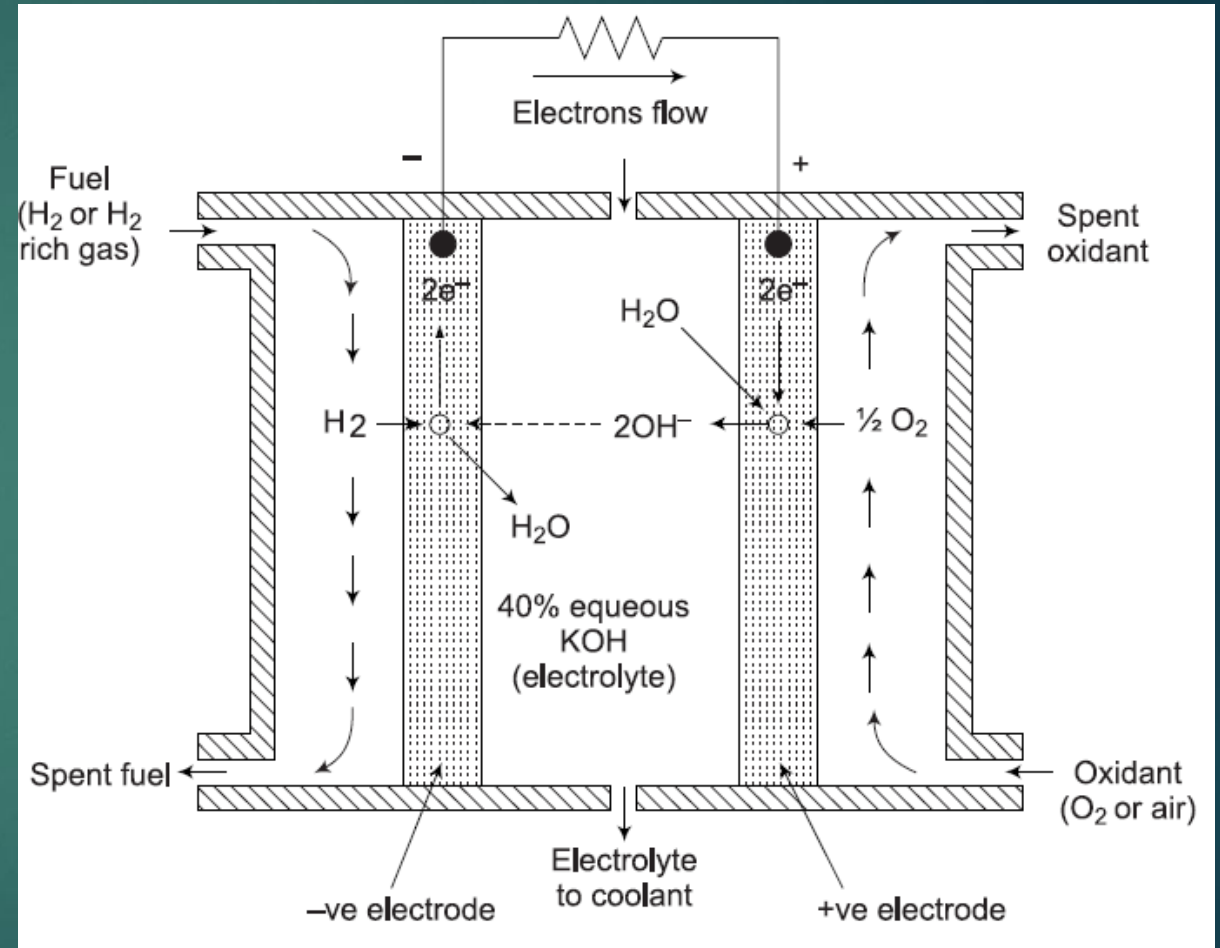
Temperature: **150–200 °C**

Ideal EMF: **1.23 V (25 °C) → 1.15 V (200 °C)**

Actual Cell Voltage: **0.7–0.8 V (rated load)**

2. Alkaline Fuel Cell (AFC)

Uses an aqueous alkaline solution as the electrolyte. These are efficient but highly sensitive to carbon dioxide, so they are often used in controlled environments like spacecraft.



2. Alkaline Fuel Cell (AFC)

Oldest type of fuel cell.

Electrolyte: 40% aqueous **KOH** solution.

Operating Temperature: ~90 °C.

Fuel/Oxidant: **Hydrogen (H₂)** and **Oxygen (O₂)** — same active materials as PAFC.

Electrodes: Similar to PAFC, porous and catalyst-coated.

EMF: Comparable to PAFC (~0.7–0.8 V).

Electrochemical Reactions:

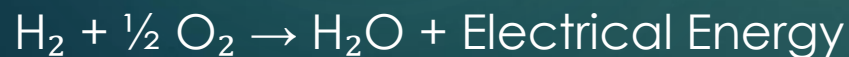
At Cathode (Positive Electrode):



At Anode (Negative Electrode):



Overall Reaction:



Features & Limitations:

Charge Carrier: OH⁻ ions move from cathode → anode.

CO₂ Sensitive: CO₂ reacts with KOH forming K₂CO₃, increasing resistance.

Requires CO₂-free Fuel & Oxidant:

Pure hydrogen and oxygen preferred.

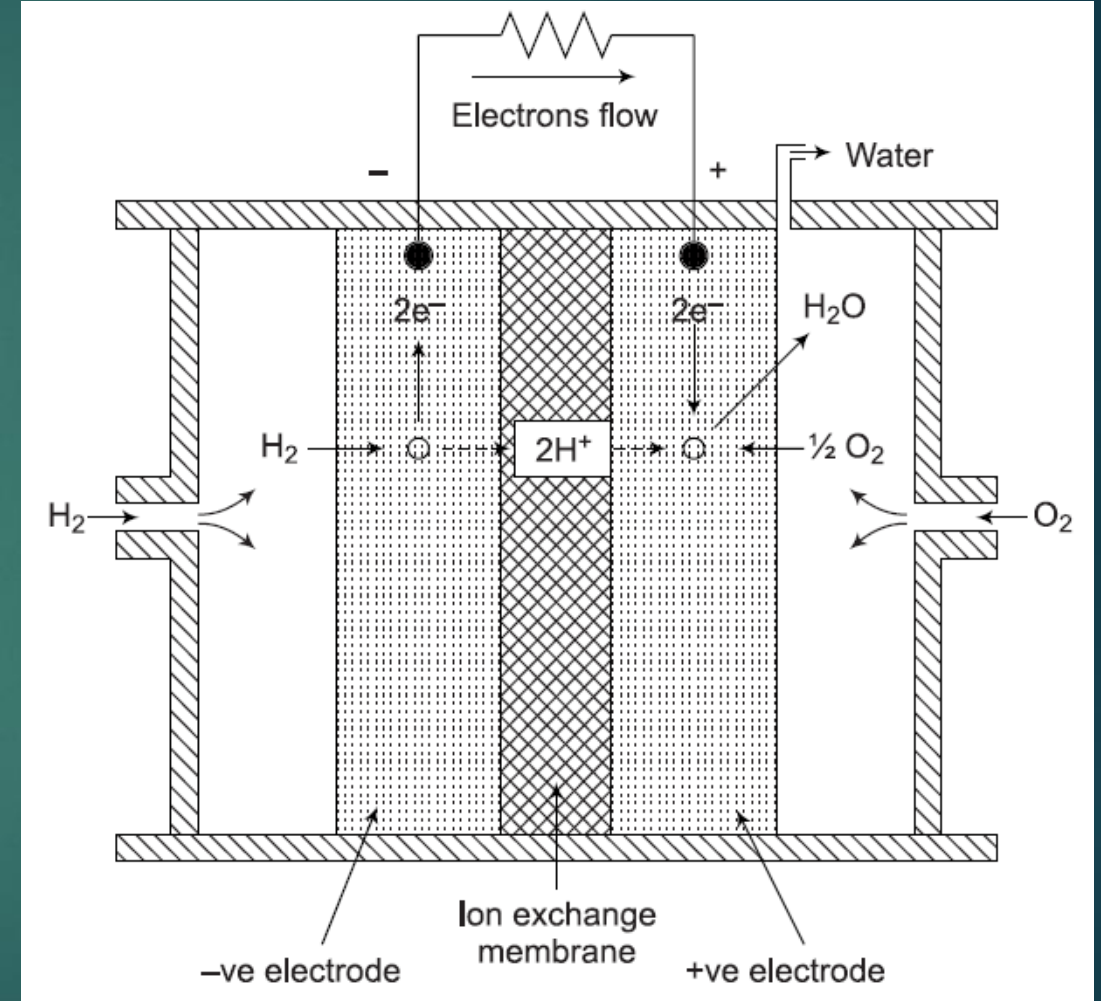
If air used → CO₂ must be **removed by lime scrubbing**.

Applications:

Used in **space missions (Apollo program)** and **military systems** due to high efficiency and reliability.

3. Polymer Electrolyte Membrane (PEMFC):

Also known as Proton Exchange Membrane fuel cells, these use a special polymer membrane that conducts protons. They operate at relatively low temperatures (50° - 100°) and are well-suited for vehicles and portable power.



3. Polymer Electrolyte Membrane (PEMFC):

Overview:

Uses a **solid polymer membrane** (e.g., *polystyrene sulphonic acid*) as electrolyte.

Electrolyte Function: Allows passage of **H⁺ ions**, blocks gases (H₂, O₂).

Operating Temperature: 40–60 °C

Electrode Catalyst: Finely divided **Platinum (Pt)** on both sides of membrane.

Membrane Thickness: ~0.076 cm (to minimize internal resistance).

Desired Membrane Properties:

High **ionic conductivity**.

Non-permeable to H₂ and O₂ gases.

Low **electro-osmosis**.

High **dehydration resistance**.

Strong **oxidation & hydrolysis resistance**.

Good **mechanical stability**.

Electrochemical Reactions:

Anode: $\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$

Cathode: $\frac{1}{2}\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2\text{O}$

Overall: $\text{H}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{H}_2\text{O} + \text{Electrical Energy}$

Features:

Coolant tubes remove heat; **wicks** manage water via capillary action.

Excess water expelled, maintaining optimal hydration.

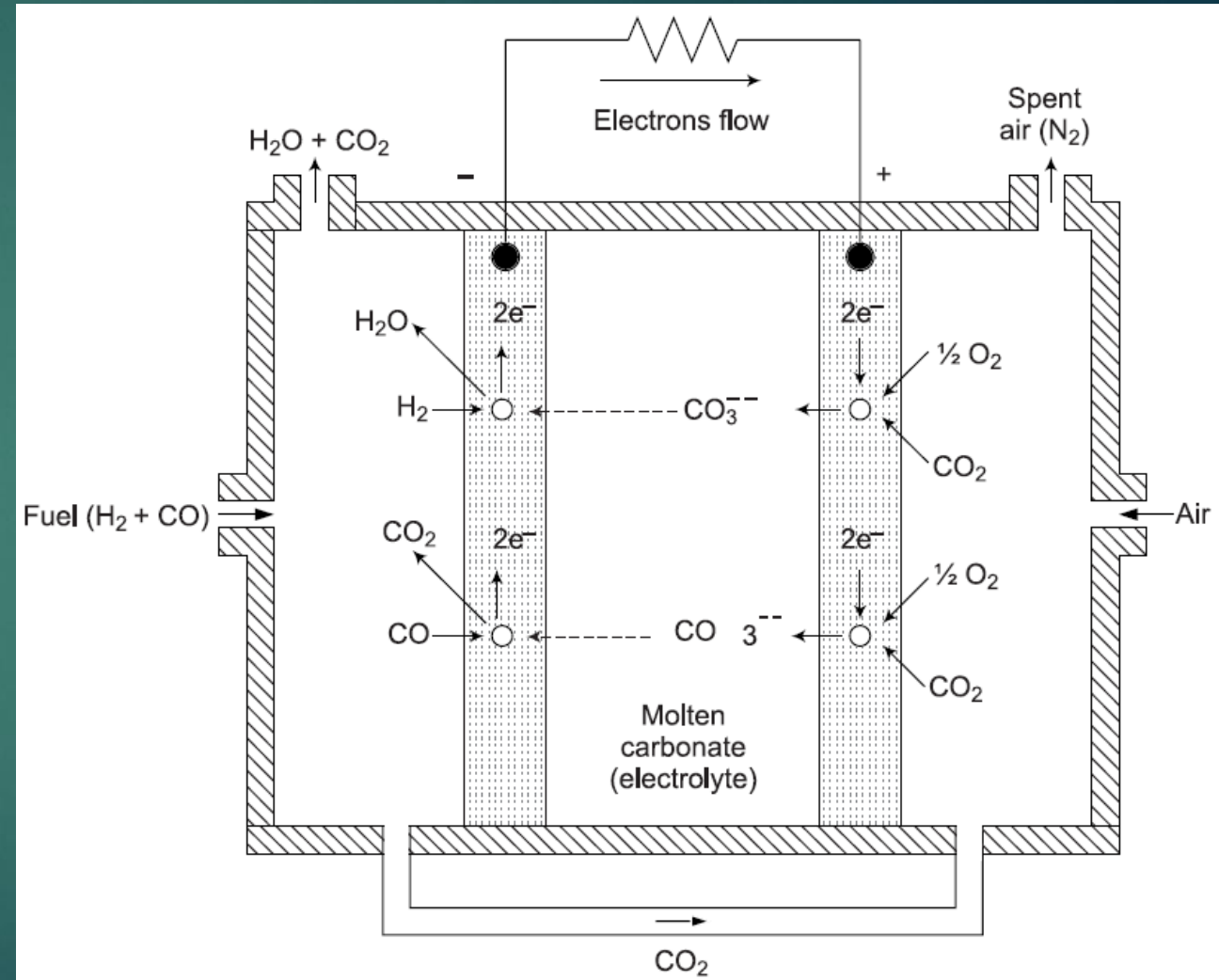
Ideal EMF: 1.23 V at 25 °C.

Applications:

Widely used in **automotive fuel cells**, **portable power**, and **spacecraft**.

4. Molten Carbonate Fuel Cell (MCFC):

Uses a molten mixture of carbonate salts as the electrolyte. These operate at high temperatures and can use fuels like natural gas, but require high-temperature materials.



4. Molten Carbonate Fuel Cell (MCFC):

Overview:

Electrolyte: Molten carbonates of **Na, K, or Li** held in a **ceramic matrix**.

Operating Temperature: **600–700 °C** (above melting point of carbonates).

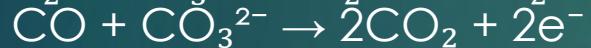
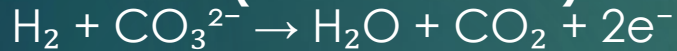
Electrodes: **Porous nickel** (no catalyst required due to high temperature).

Fuel: **H₂** and **CO** (synthesis gas); derived from reforming fossil fuels like coal or natural gas.

EMF: Theoretical $\approx 1 \text{ V}$ at 700 °C; **Actual $\approx 0.8 \text{ V}$** under load.

Electrochemical Reactions:

At Anode (Fuel Electrode):



At Cathode (Oxidant Electrode):



Overall Reaction:



Charge Carrier:

Carbonate ions (CO₃²⁻) move from **cathode** → **anode** through electrolyte.

Features & Advantages:

Operates on **low-cost fuels** (H₂ + CO).

High exhaust temperature (>540 °C) allows **waste heat recovery** for steam or process heat.

Enables **high overall efficiency** with combined heat and power (CHP) systems.

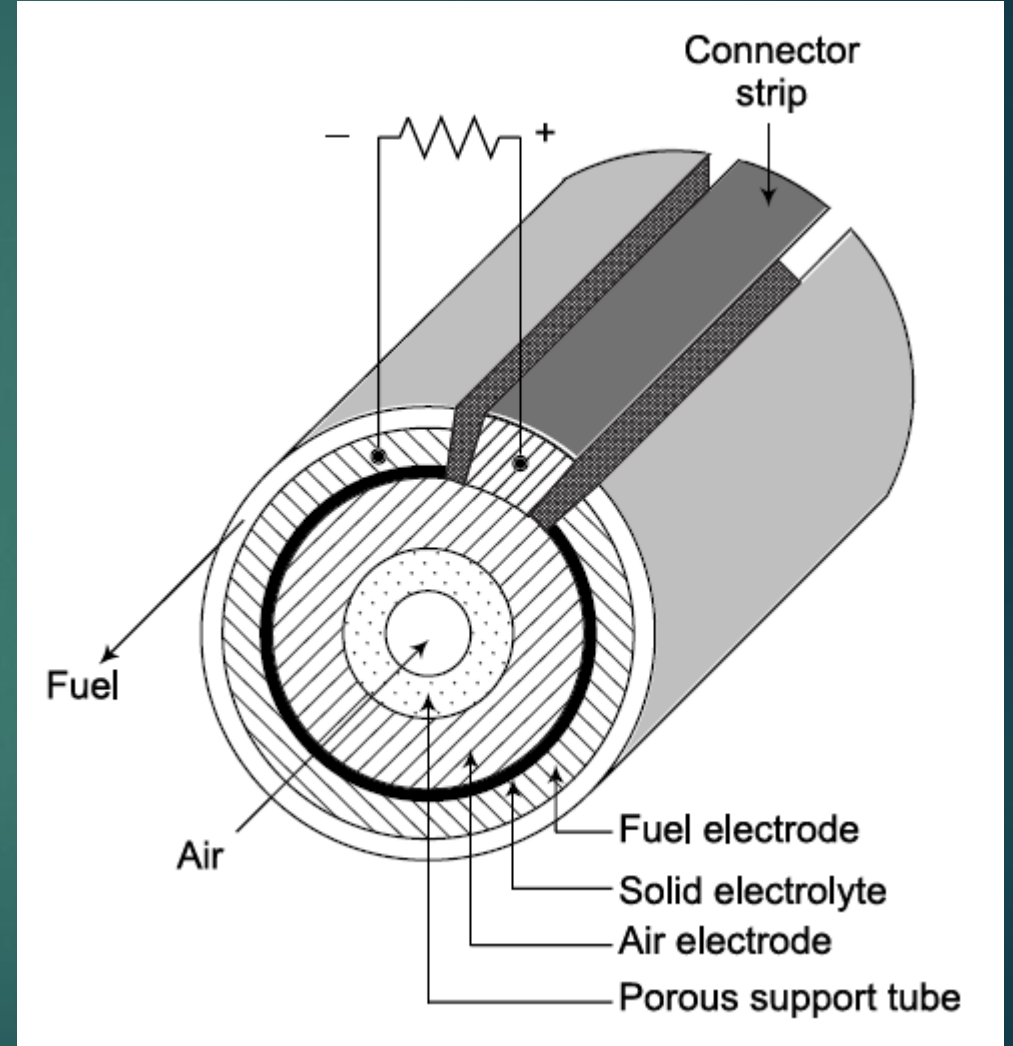
No precious metal catalysts required → **lower cost**.

Applications:

Industrial power plants and **combined-cycle systems** using coal or natural gas.

5. Solid Oxide Fuel Cell (SOFC):

Uses a solid ceramic material as the electrolyte. SOFCs operate at very high temperatures, making them less sensitive to fuel impurities and suitable for stationary power generation, but they have long startup times.



Overview:

Electrolyte: Solid oxide (ceramic) — typically **stabilized zirconium oxide (ZrO₂)**.

Conducts **oxide ions (O²⁻)** at high temperature.

Operating Temperature: 600–1000 °C.

Electrodes:

Anode (Fuel Electrode): Porous **nickel**.

Cathode (Oxidant Electrode): **Metal oxide** (e.g., indium oxide).

Catalyst: *Not required* due to high operating temperature.

Fuel: H₂ and CO — same as in MCFC.

Typical Output Voltage: ~**0.63 V** under load.

Electrochemical Reactions:

At Anode (Fuel Electrode):



At Cathode (Oxidant Electrode):



Overall Reaction:



Key Features & Advantages:

- **Charge Carrier:** O²⁻ ions move from **cathode → anode**.
- **High efficiency** and potential for **combined heat and power (CHP)** systems.
- **Fuel flexibility:** Can operate on hydrogen, carbon monoxide, or reformed hydrocarbons.
- **Tubular design** improves strength and thermal management.
- **Waste heat utilization** for additional power or industrial use.

Applications:

- **Stationary power generation, industrial co-generation, and auxiliary power units (APUs).**